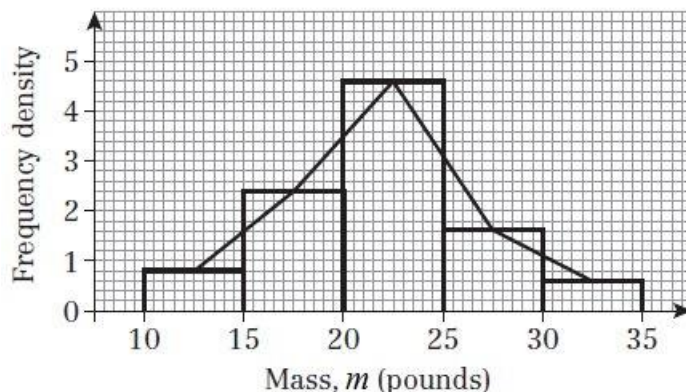


Representations of data 3D

- 1 a Class widths are all 5.

Frequency densities are: 0.8, 2.4, 4.6, 1.6, 0.6



- 2 a Time is a continuous variable.

- b 90 students took between 60 and 70 seconds

Frequency = frequency density $\times$ class width $\times k$

$$90 = 6 \times 10 \times k$$

$$k = 1.5$$

For the number of students who took between 40 and 60 seconds:

$$5 \times 20 \times 1.5 = 150$$

So there were 150 students who took between 40 and 60 seconds.

- c 150 students took between 40 and 60 seconds (from part b)

$$6 \times 10 \times 1.5 = 90 \text{ students took between 60 and 70 seconds.}$$

$$8.6 \times 10 \times 1.5 = 129 \text{ students took between 70 and 80 seconds.}$$

So $150 + 90 + 129 = 369$ students took 80 seconds or less.

- d 369 students took 80 seconds or less (from part c)

$$4 \times 5 \times 1.5 = 105 \text{ students took between 80 and 85 seconds.}$$

$$12 \times 5 \times 1.5 = 90 \text{ students took between 85 and 90 seconds.}$$

$$3 \times 30 \times 1.5 = 135 \text{ students took between 90 and 120 seconds.}$$

$$\text{Total: } 369 + 105 + 90 + 135 = 699$$

There are 699 students in total.

- 3 a Distance is a continuous variable.

- b Frequency = frequency density $\times$ class width

$$0 \text{ m to } 20 \text{ m: } 2 \times 20 = 40$$

$$20 \text{ m to } 35 \text{ m: } 5 \times 15 = 75$$

$$35 \text{ m to } 45 \text{ m: } 10 \times 10 = 100$$

$$45 \text{ m to } 60 \text{ m: } 6 \times 15 = 90$$

$$60 \text{ m to } 65 \text{ m: } 1 \times 5 = 5$$

$$\text{Total: } 40 + 75 + 100 + 90 + 5 = 310$$

So 310 people entered the competition.

- 3 c** Frequency = frequency density $\times$ class width

30 m to 35 m: $5 \times 5 = 25$

35 m to 40 m: $10 \times 5 = 50$

Total: $25 + 50 = 75$

So 75 people threw between 30 and 40 metres.

- 3 d** From part **b**:

45 m to 60 m: 90 people

60 m to 65 m: 5 people

Total: $90 + 5 = 95$

So 95 people threw between 45 and 60 metres.

- e** From part **b**, 40 people threw between 0 and 20 metres.

20 m to 25 m: $5 \times 5 = 25$

Total: $40 + 25 = 65$

So 65 people threw between 0 and 25 metres.

- 4 a** The bar for $28 \leq m < 32$ has an area of $10 \times 10 = 100$ squares.

If 100 squares represents 32 lambs then

$$\frac{100}{4} \text{ squares represents } \frac{32}{4} \text{ lambs.}$$

i.e. 25 squares represents 8 lambs.

- b** The class $24 \leq m < 26$ contains $5 \times 20 = 100$ squares.

As above, this represents 32 lambs.

- c** The class $20 \leq m < 24$ contains $10 \times 10 = 100$ squares which represents 32 lambs.

The class $24 \leq m < 26$ contains $5 \times 20 = 100$ squares which represents 32 lambs.

The class $26 \leq m < 28$ contains $5 \times 40 = 200$ squares which represents 64 lambs.

The class $28 \leq m < 32$ contains $10 \times 10 = 100$ squares which represents 32 lambs.

The class $32 \leq m < 34$ contains $5 \times 5 = 25$ squares which represents 8 lambs.

So in total we have $32 + 32 + 64 + 32 + 8 = 168$ lambs.

- d** Class $25 \leq m < 26$ is approximately $\frac{1}{2}$ of class $24 \leq m < 26$ which equates to 16 lambs.

Class $26 \leq m < 28$ represents 64 lambs.

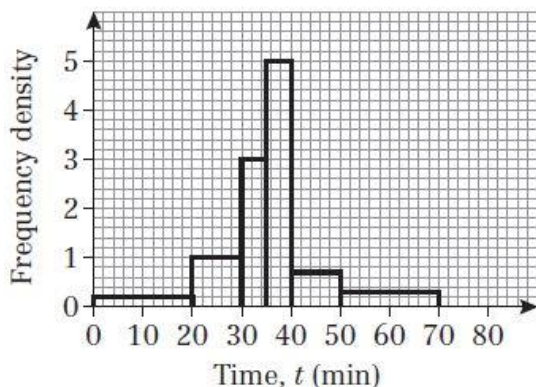
Class $28 \leq m < 29$ is approximately $\frac{1}{4}$ of class $28 \leq m < 32$ which equates to 8 lambs.

So in total we have $16 + 64 + 8 = 88$ lambs.

- 5 a i Use extra columns to help, using the frequency densities given in the histogram:

Time, t (min)	Frequency	Class width	Frequency density
$0 \leq t < 20$	4	20	0.2
$20 \leq t < 30$	$10 \times 1 = 10$	10	1
$30 \leq t < 35$	15	5	3
$35 \leq t < 40$	25	5	5
$40 \leq t < 50$	$10 \times 0.7 = 7$	10	0.7
$50 \leq t < 70$	$20 \times 0.3 = 6$	20	0.3

- 5 a ii



b $\left(\frac{5}{10} \times 10\right) + 15 + \left(\frac{3}{5} \times 25\right) = 35$ passengers.

- 6 a 12.5 and 14.5 are the class boundaries, as we are dealing with continuous data.

- b i The class boundaries for the 15–17 class are 14.5 and 17.5.
This width is 1.5 times the width of the 13–14 class, since $17.5 - 14.5 = 3 = 1.5 \times 2$.
So the width of the class is $1.5 \times 4 = 6$ cm.

- ii The frequency density for the 13–14 class is $\frac{24}{2} = 12$.

The frequency density of this class is 6, which is 0.5 times the frequency density above: 12.
So the height of the class is $0.5 \times 6 = 3$ cm.

- 7 a Width is half of the $8 \leq t < 10$ class, which is 0.5 cm.

Height is double the frequency density, so must be $\frac{7}{1} \times 2 = 14$ cm.

b Mean = $\frac{\sum fx}{\sum f} = \frac{645}{62} = 10.4^\circ\text{C}$ (to 3 s.f.) where x is taken as the midpoint of each class.

Standard deviation = $\sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2} = \sqrt{113.89 - \left(\frac{645}{62}\right)^2} = 2.4$ (to 2 s.f.)

c Q_1 is the $\frac{31}{4} = 7.75^{\text{th}}$ piece of data.

$$\frac{Q_1 - 8}{10 - 8} = \frac{7.75 - 4}{12 - 4} \text{ using linear interpolation on the } 8 \leq t < 10 \text{ class.}$$

$$Q_1 = 8.94^\circ\text{C}$$

d Mean + standard deviation = 12.8°C

$$\frac{12.8 - 12}{15 - 12} = \frac{d - 25}{30 - 25}$$

$$d = 26.33 \text{ days}$$

So 4.7 days